

DATABASE_SCHEMA

HelixTrack Database Schema — 2026-06-21

Overview

HelixTrack uses SQLite (development) or PostgreSQL (production) with 71 tables across V1-V3.

V1 Core Tables (57)

Business Entities

Table	Purpose	Status
project	Projects/workspaces	✅ Implemented
ticket	Issues/tasks	✅ Implemented
user	Users	✅ Implemented
comment	Comments	✅ Implemented
team	Teams	✅ Implemented
attachment	File attachments	⚠️ Partial
label	Labels/tags	❌ Missing
priority	Priorities	❌ Missing
resolution	Resolutions	❌ Missing
component	Components	❌ Missing
cycle	Sprints/iterations	❌ Missing

Workflow Entities

Table	Purpose	Status
workflow	Workflow definitions	❌ Missing
workflow_step	Workflow steps	❌ Missing

Table	Purpose	Status
ticket_type	Ticket types	✗ Missing
ticket_status	Ticket statuses	✗ Missing

Board Entities

Table	Purpose	Status
board	Kanban/Scrum boards	✗ Missing
board_column	Board columns	✗ Missing
board_ticket	Board-ticket mapping	✗ Missing

User Entities

Table	Purpose	Status
user_role	User roles	✗ Missing
permission	Permissions	✗ Missing
user_preference	User preferences	✗ Missing

System Entities

Table	Purpose	Status
system_info	System metadata	✗ Missing
audit_log	Audit trail	✗ Missing
notification	Notifications	✗ Missing

V2 Extension Tables (14)

Enhanced Features

Table	Purpose	Status
ticket_relationship	Ticket relationships	⚠ Partial
ticket_link	Ticket links	⚠ Partial
ticket_watchers	Ticket watchers	⚠ Partial
ticket_time_entry	Time tracking	⚠ Partial

Workflow Extensions

Table	Purpose	Status
workflow_transition	Workflow transitions	✗ Missing

Table	Purpose	Status
workflow_condition	Workflow conditions	✗ Missing
workflow_action	Workflow actions	✗ Missing

V3 Advanced Tables

Security

Table	Purpose	Status
security_audit	Security audit log	✓ Implemented
permission_cache	Permission cache	✓ Implemented
account	Account management	✓ Implemented

Table Relationships

Core Relationships

project (1) — (N) ticket
 project (1) — (N) board
 project (1) — (N) cycle
 project (1) — (N) component
 ticket (1) — (N) comment
 ticket (1) — (N) attachment
 ticket (N) — (N) label
 user (1) — (N) ticket (assignee)
 user (1) — (N) comment
 team (N) — (N) user

Indexes

Performance Indexes

- idx_ticket_project_id — Ticket lookup by project
- idx_ticket_status — Ticket lookup by status
- idx_ticket_assignee — Ticket lookup by assignee
- idx_comment_ticket_id — Comment lookup by ticket
- idx_audit_created — Audit log by date
- idx_audit_user_id — Audit log by user

Migrations

Migration History

- V1.0 — Initial schema
- V2.0 — Extended features
- V3.0 — Advanced features
- V5.6 — Security engine enhancement

Migration Files

- `core/Application/Database/DDL/Migration.V5.6.sql`
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Cross-references

- [Architecture](#)
- [API Reference](#)

Sources verified 2026-06-22: /Volumes/T7/Projects/helix_track/core/Database/DDL/Definition.V1..V5.sql , /Volumes/T7/Projects/helix_track/core/Application/Database/DDL/ , <https://github.com/Helix-Track/Everything>

REPO-STATE-DERIVED (per §11.4.99 the sources are the cross-referenced DDL files, following the docs/ARCHITECTURE.md precedent). Cross-referenced on 2026-06-22: - **Schema DDL CONFIRMED present:** helix_track/core/Database/DDL/Definition.V1.sql ... Definition.V5.sql exist (plus Indexes_Performance.sql, Migration.V*.sql), and the migration set also lives at core/Application/Database/DDL/ (Migration.V5.6.sql, Test_Data_Users_Permissions.sql) — the doc's core/Application/Database/DDL/... migration path is real. SQLite (dev) source Definition.sqlite is present. - **Negative finding (version-range drift):** this doc states “71 tables across **V1-V3**”, but the live DDL tree carries Definition files **V1 through V5** (and a V5.6 migration). The table count and the “V1-V3” range should be re-derived from the V1-V5 DDL on the next revision — the doc's range understates the shipped schema versions. Re-verify table counts directly from the Definition.V*.sql DDL, which is the source of truth.